2012 Mathematics

Higher

Finalised Marking Instructions

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General Comments

These marking instructions are for use with the 2012 Higher Mathematics Examination.

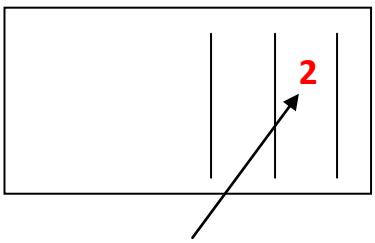
For each question the marking instructions are in two sections, namely **Illustrative Scheme** and **Generic Scheme**. The **Illustrative Scheme** covers methods which are commonly seen throughout the marking. The **Generic Scheme** indicates the rationale for which each mark is awarded. In general markers should use the **Illustrative Scheme** and only use the **Generic Scheme** where a candidate has used a method not covered in the **Illustrative Scheme**.

All markers should apply the following general marking principles throughout their marking:

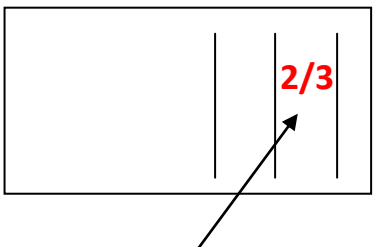
1 Marks must be assigned in accordance with these marking instructions. In principle, marks are awarded for what is correct, rather than deducted for what is wrong.

2 Award one mark for each •. There are no half marks.

3 The mark awarded for **each part** of a question should be entered in the **outer** right hand margin, opposite the end of the working concerned. The marks should correspond to those on the question paper and these marking instructions. Only the mark, **as a whole number**, should be written.



Marks in this column - whole numbers only



Do not record marks on scripts in this manner.

4 Where a candidate has not been awarded any marks for a question, or part of a question, 0 should be written in the right hand margin against their answer. It should not be left blank.

5 Every page of a candidate's script should be checked for working. Unless blank, every page which is devoid of a marking symbol should have a tick placed in the bottom right hand margin.

6 Where the solution to part of a question is fragmented and continues later in the script, the marks should be recorded at the end of the solution. This should be indicated with a down arrow (↓), in the margin, at the earlier stages.

7 Working subsequent to an error must be **followed through**, with possible full marks for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working for a follow through mark has been eased, the follow through mark cannot be awarded.

8 As indicated on the front of the question paper, full credit should only be given where the solution contains appropriate working. Throughout this paper, unless specifically mentioned in the marking instructions, a correct answer with no working receives no credit.

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9 Marking Symbols

No comments or words should be written on scripts. Please use the following and the symbols indicated on the welcome letter and from comment 6 on the previous page.

✓ A tick should be used where a piece of working is correct and gains a mark. Markers must check through the whole of a response, ticking the work only where a mark is awarded.

—— X At the point where an error occurs, the error should be underlined and a cross used to indicate where a mark has not been awarded. If no mark is lost the error should only be underlined, i.e. a cross is only used where a mark is not awarded.

✓✗ A cross-tick should be used to indicate "correct" working where a mark is awarded as a result of **follow through** from an error.

✓✗ A double cross-tick should be used to indicate correct working which is irrelevant or insufficient to score any marks. This should also be used for working which has been **eased**.

~~~ A tilde should be used to indicate a minor error which is not being penalised, e.g. **bad form**.

✓~~~ This should be used where a candidate is given the **benefit of the doubt**.

^ A roof should be used to show that something is missing, such as part of a solution or a crucial step in the working.

These will help markers to maintain consistency in their marking and will assist the examiners in the later stages of SQA procedures.

The examples below illustrate the use of the marking symbols .

Example 1

$y = x^3 - 6x^2$

$\frac{dy}{dx} = 3x^2 - 12$ ✓ X

$3x^2 - 12 = 0$ ✓✗

$x = 2$ ^

$y = -16$ ✓✗

$\bullet^1$ ✓

$\bullet^2$ X

$\bullet^3$ ✓✗

$\bullet^4$ ^

$\bullet^5$ ✓✗

Example 2

$A(4,4,0),\ B(2,2,6),\ C(2,2,0)$

$\overrightarrow{AB} = \mathbf{b} + \mathbf{a} = \begin{pmatrix} 6 \\ 6 \\ 6 \end{pmatrix}$ X $\bullet^1$

$\overrightarrow{AC} = \begin{pmatrix} 6 \\ 6 \\ 0 \end{pmatrix}$ ✓✗ $\bullet^2$ (repeated error)

Example 3

$3\sin x - 5\cos x$

$k\sin x\cos a - \cos x\sin a$ ✓ $\bullet^1$

$k\cos a = 3,\ k\sin a = 5$ ✓ $\bullet^2$

Example 4

$$\begin{array}{c|cccc} 4 & 1 & -5 & 2 & 8 \\ & & 4 & -4 & -8 \\ \hline & 1 & 1 & -2 & 0 \end{array}$$ ✓ $\bullet^1$ ✓ $\bullet^2$

Since the remainder is 0, $x-4$ must be a factor. ✓ $\bullet^3$

$(x^2 - x - 2)$ ✓ $\bullet^4$

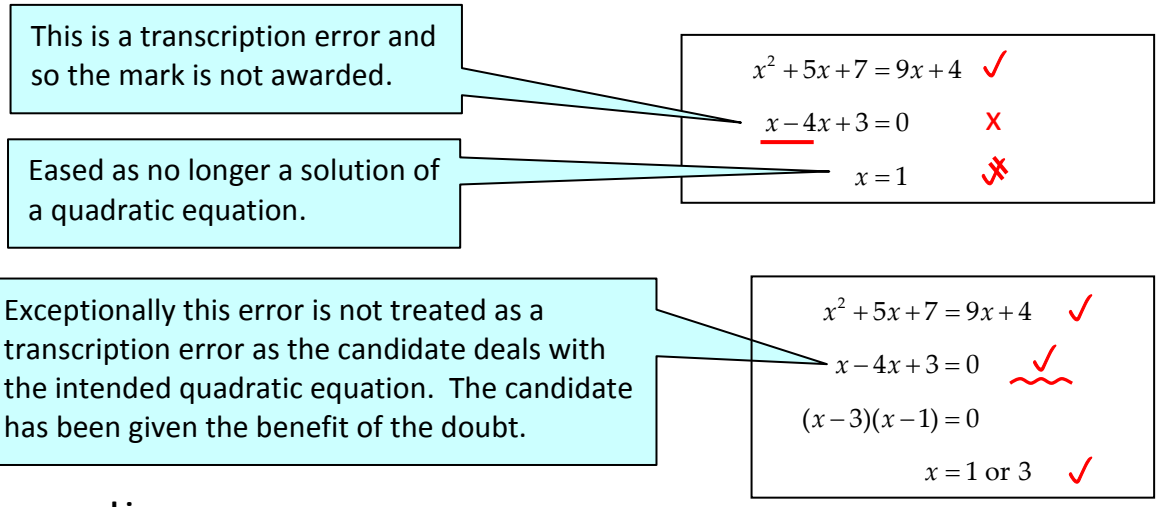
$(x-4)(x+1)(x-2)$ ✓ $\bullet^5$

$x = 4$ or $x = -1$ or $x = 2$ ✓ $\bullet^6$

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~~~

10 In general, as a consequence of an error perceived to be trivial, casual or insignificant, e.g. $6 \times 6 = 12$, candidates lose the opportunity of gaining a mark. But note example 4 in comment 9 and comment 11.

11 Where a transcription error (paper to script or within script) occurs, the candidate should be penalised, e.g.



12 **Cross marking**

Where a question results in two pairs of solutions, this technique should be applied, but only if indicated in the detailed marking instructions for the question.

Example: Point of intersection of line with curve

Illustrative Scheme: $\bullet^5$ $x = 2, x = -4$ Cross marked: $\bullet^5$ $x = 2, y = 5$

$\bullet^6$ $y = 5, y = -7$ $\bullet^6$ $x = -4, y = -7$

Markers should choose whichever method benefits the candidate, but **not** a combination of both.

13 In final answers, numerical values should be simplified as far as possible.

Examples: $\frac{15}{12}$ must be simplified to $\frac{5}{4}$ or $1\frac{1}{4}$ $\frac{43}{1}$ must be simplified to 43

$\frac{15}{0{\cdot}3}$ must be simplified to 50 $\frac{4/5}{3}$ must be simplified to $\frac{4}{15}$

$\sqrt{64}$ must be simplified to 8

The square root of perfect squares up to and including 100 must be known.

14 Regularly occurring responses (ROR) are shown in the marking instructions to help mark common and/or non-routine solutions. RORs may also be used as a guide in marking similar non-routine candidate responses.

15 Unless specifically mentioned in the marking instructions, the following should not be penalised:

- Working subsequent to a **correct** answer;
- Correct working in the wrong part of a question;
- Legitimate variations in numerical answers, e.g. angles in degrees rounded to nearest degree;
- Omission of units;
- Bad form;
- Repeated error within a question, but not between questions or papers.

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16 In any 'Show that . . .' question, where the candidate has to arrive at a formula, the last mark of that part is not available as a follow through from a previous error.

17 All working should be carefully checked, even where a fundamental misunderstanding is apparent early in the candidate's response. Marks may still be available later in the question so reference must be made continually to the marking instructions. All working must be checked: the appearance of the correct answer does not necessarily indicate that the candidate has gained all the available marks.

18 In the **exceptional** circumstance where you are in doubt whether a mark should or should not be awarded, consult your Team Leader (TL).

19 Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.

20 A valid approach, within Mathematical problem solving, is to try different strategies. Where this occurs, all working should be marked. The mark awarded to the candidate is from the **highest** scoring strategy. This is distinctly different from the candidate who gives two or more solutions to a question/part of a question, deliberately leaving all solutions, hoping to gain some benefit. All such contradictory responses should be marked and the **lowest** mark given.

21 It is of great importance that the utmost care should be exercised in totalling the marks.

The recommended procedure is as follows:

Step 1 Manually calculate the total from the candidate's script.

Step 2 Check this total using the grid issued with these marking instructions.

Step 3 In EMC, enter the marks and obtain a total, which should now be compared to the manual total.

This procedure enables markers to identify and rectify any errors in data entry before submitting each candidate's marks.

22 The candidate's script for Paper 2 should be placed inside the script for Paper 1, and the candidate's total score (i.e. Paper 1 Section B + Paper 2) written in the space provided on the front cover of the script for Paper 1.

23 In cases of difficulty, covered neither in detail nor in principle in these instructions, markers should contact their TL in the first instance. A referral to the Principal Assessor (PA) should only be made in consultation with the TL. Further details of PA Referrals can be found in The General Marking Instructions.

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Paper 1 Section A

Question	Answer
1	C
2	D
3	B
4	B
5	A
6	C
7	A
8	C
9	A
10	B
11	D
12	B
13	D
14	A
15	D
16	C
17	D
18	B
19	B
20	A

Summary		
	A	5
	B	6
	C	4
	D	5

21 (a) (i) Show that $(x-4)$ is a factor of x^3-5x^2+2x+8.

(ii) Factorise x^3-5x^2+2x+8 fully.

(iii) Solve $x^3-5x^2+2x+8=0$. **6**

Generic Scheme

21 (a)

- $\bullet^1$ ss know to use $x=4$
- $\bullet^2$ pd complete evaluation
- $\bullet^3$ ic state conclusion
- $\bullet^4$ ic find quadratic factor
- $\bullet^5$ pd factorise completely
- $\bullet^6$ ic state solutions

Illustrative Scheme

Method 1 : Using synthetic division

$\bullet^1$

4	1	−5	2	8

$\bullet^2$

4	1	−5	2	8
		4	−4	−8
	1	−1	−2	0

$\bullet^3$ 'remainder is zero so $(x-4)$ is a factor'

$\bullet^4$ x^2-x-2 **stated, or implied by $\bullet^5$**

$\bullet^5$ $(x-4)(x-2)(x+1)$ **stated explicitly in any order**

$\bullet^6$ $-1,\ 2,\ 4$

Method 2 : Using substitution and inspection

$\bullet^1$ know to use $x=4$

$\bullet^2$ $64-80+8+8=0$

$\bullet^3$ $(x-4)$ is a factor

$\bullet^4$ $(x-4)(x^2-x-2)$ **stated, or implied by $\bullet^5$**

$\bullet^5$ $(x-4)(x-2)(x+1)$ **stated explicitly in any order**

$\bullet^6$ $-1,\ 2,\ 4$

6

Notes

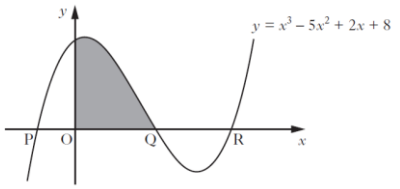
1. $\bullet^3$ is only available as a consequence of the evidence for $\bullet^1$ and $\bullet^2$.
2. Communication at $\bullet^3$ must be consistent with working at $\bullet^2$.
   i.e. candidate's working must arrive legitimately at zero before $\bullet^3$ is awarded.
   If the remainder is not 0 then an appropriate statement would be '$(x-4)$ is not a factor'.
3. Accept any of the following for $\bullet^3$:
   - '$f(4)=0$ so $(x-4)$ is a factor'
   - 'since remainder is 0, it is a factor'
   - the 0 from table linked to word 'factor' by e.g. 'so', 'hence', '$\therefore$', '$\rightarrow$', '$\Rightarrow$'.
4. Do not accept any of the following for $\bullet^3$:
   - double underlining the zero or boxing in the zero, without a comment
   - '$x=4$ is a factor', '$(x+4)$ is a factor', '$x=4$ is a root', '$(x-4)$ is a root'
   - the word 'factor' **only**, with no link.
5. To gain $\bullet^6$, $4, -1, 2$ **must** appear together in (a).
6. $(x-4)(x-2)(x+1)$ leading to $(4,\ 0)$, $(2,\ 0)$ and $(-1,\ 0)$ **only** does not gain $\bullet^6$.
7. $(x-2)(x+1)$ only, leading to $x=2$, $x=-1$ does not gain $\bullet^6$ as equation solved is not a cubic.
8. Candidates who attempt to solve the cubic equation subsequent to $x=-1, 2, 4$ and obtain different solutions, or no solutions, cannot gain $\bullet^6$.

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21 (b) The diagram shows the curve with equation $y = x^3 - 5x^2 + 2x + 8$.

The curve crosses the x-axis at P, Q and R.

Determine the shaded area.



6

Generic Scheme

21 (b)

- $\bullet^7$ ic identify x_Q from working in (a)
- $\bullet^8$ ic interpret appropriate limits
- $\bullet^9$ ss know and start to integrate
- $\bullet^{10}$ pd complete integration
- $\bullet^{11}$ ic substitute limits
- $\bullet^{12}$ pd state area

Illustrative Scheme

- $\bullet^7$ 2
- $\bullet^8$ 0, 2
- $\bullet^9$ integrate one term correctly (but see Note 10)
- $\bullet^{10}$ $\frac{1}{4}x^4 - \frac{5}{3}x^3 + \frac{2}{2}x^2 + 8x$ or equivalent
- $\bullet^{11}$ $\left(\frac{1}{4}(2)^4 - \frac{5}{3}(2)^3 + 2^2 + 8\times 2\right) - 0$
- $\bullet^{12}$ $\frac{32}{3}$ or $10\frac{2}{3}$ but not a decimal approximation

6

Notes

9. Evidence for $\bullet^7$ and $\bullet^8$ may not appear until $\bullet^{11}$ stage.
10. Where a candidate differentiates one or more terms at $\bullet^9$, then $\bullet^9$, $\bullet^{10}$, $\bullet^{11}$ and $\bullet^{12}$ are not available.
11. Candidates who substitute at $\bullet^{11}$, without integrating at $\bullet^9$, do not gain $\bullet^9$, $\bullet^{10}$, $\bullet^{11}$ and $\bullet^{12}$.
12. For candidates who make an error in (a), $\bullet^8$ is only available if 0 is the lower limit and a positive integer value is used for the upper limit.
13. $\bullet^{11}$ is only available where both limits are numerical values.
14. Candidates must show evidence that they have considered the lower limit 0 in their substitution at $\bullet^{11}$ stage.

Regularly occurring responses

Response 1

Candidates who use Q throughout

Candidate A

$$\int_0^Q (x^3 - 5x^2 + 2x + 8)\,dx \qquad \bullet^7\ \text{X}$$

$$= \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + \frac{2}{2}x^2 + 8x\right]_0^Q \qquad \bullet^8\ \text{X},\ \bullet^9\ \checkmark$$

$$= \frac{1}{4}Q^4 - \frac{5}{3}Q^3 + Q^2 + 8Q - 0 \qquad \bullet^{10}\ \checkmark,\ \bullet^{11}\ \text{X},\ \bullet^{12}\ \text{X}$$

However, if Q is replaced by 2 at this stage, and working continues, all 6 marks may still be available .

Response 2

Dealing with negatives

Candidate B

Q(−1, 0) X $\bullet^7$

$$\int_0^{-1} (x^3 - 5x^2 + 2x + 8)\,dx \ \text{X}\ \bullet^8$$

$$= \left[\frac{1}{4}x^4 - \frac{5}{3}x^3 + x^2 + 8x\right]_0^{-1} \qquad \checkmark\bullet^9,\ \checkmark\bullet^{10}$$

$$= \frac{1}{4}(-1)^4 - \frac{5}{3}(-1)^3 + (-1)^2 + 8(-1) - 0 \ \checkmark\text{X}\ \bullet^{11}$$

$$= -\frac{61}{12}$$

cannot be negative so $\frac{61}{12}$ X $\bullet^{12}$

but

$$= -\frac{61}{12}$$

$$\text{A} = \frac{61}{12} \ \checkmark\text{X}\ \bullet^{12}$$

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22 (a) The expression $\cos x - \sqrt{3}\sin x$ can be written in the form $k\cos(x+a)$ where $k>0$ and $0 \le a < 2\pi$.
Calculate the values of k and a. **4**

Generic Scheme

22 (a)

- $\bullet^1$ ss use compound angle formula
- $\bullet^2$ ic compare coefficients
- $\bullet^3$ pd process k
- $\bullet^4$ pd process a

Illustrative Scheme

- $\bullet^1$ $k\cos x\cos a - k\sin x\sin a$ **stated explicitly**
- $\bullet^2$ $k\cos a = 1$ and $k\sin a = \sqrt{3}$ **stated explicitly**
- $\bullet^3$ 2 (do not accept $\sqrt{4}$)
- $\bullet^4$ $\frac{\pi}{3}$ **but** must be consistent with $\bullet^2$

4

Notes

1. Treat $k\cos x\cos a - \sin x\sin a$ as bad form only if the equations at the $\bullet^2$ stage both contain k.
2. $2\cos x\cos a - 2\sin x\sin a$ or $2(\cos x\cos a - \sin x\sin a)$ is acceptable for $\bullet^1$ and $\bullet^3$.
3. Accept $k\cos a = 1$ and $-k\sin a = -\sqrt{3}$ for $\bullet^2$.
4. $\bullet^2$ is not available for $k\cos x = 1$ and $k\sin x = \sqrt{3}$, however, $\bullet^4$ is still available.
5. $\bullet^4$ is only available for a single value of a.
6. Candidates who work in degrees and do not convert to radian measure in (a) do not gain $\bullet^4$.
7. Candidates may use any form of the wave equation for $\bullet^1$, $\bullet^2$ and $\bullet^3$, however, $\bullet^4$ is only available if the value of a is interpreted for the form $k\cos(x+a)$.

Regularly occurring responses

Response 1 : Missing information in working

Candidate A

$2\cos a = 1$ ✓
$-2\sin a = -\sqrt{3}$ ✓
$\tan a = \frac{\sqrt{3}}{1}$
$a = \frac{\pi}{3}$ ✓

$\bullet^1$ ✗
$\bullet^2$ ✓
$\bullet^3$ ✓
$\bullet^4$ ✓

3 marks out of 4

Candidate B

$\cos a = 1$
$\sin a = \sqrt{3}$
$\tan a = \frac{\sqrt{3}}{1}$
$a = \frac{\pi}{3}$

$\bullet^1$ ✗
$\bullet^2$ ✗
$\bullet^3$ ✗
$\bullet^4$ ✗

0 marks out of 4

Not consistent with evidence at $\bullet^2$.

Response 2 : Correct expansion of $k\cos(x+a)$ and possible errors for $\bullet^2$ and $\bullet^4$

Candidate C

$k\cos a = 1$
$k\sin a = \sqrt{3}$ ✓ $\bullet^2$
$\tan a = \frac{1}{\sqrt{3}}$ so $a = \frac{\pi}{6}$ ✗ $\bullet^4$

Candidate D

$k\cos a = \sqrt{3}$ ✗ $\bullet^2$
$k\sin a = 1$
$\tan a = \frac{1}{\sqrt{3}}$ so $a = \frac{\pi}{6}$ ✓ $\bullet^4$

Candidate E

$k\cos a = 1$
$k\sin a = -\sqrt{3}$ ✗ $\bullet^2$
$\tan a = -\sqrt{3}$ so $a = \frac{5\pi}{3}$ ✓ $\bullet^4$

Response 3 : Labelling incorrect using $\cos(A+B) = \cos A\cos B - \sin A\sin B$ from formula list

Candidate F

$k\cos A\cos B - k\sin A\sin B$ ✗ $\bullet^1$
$k\cos a = 1$
$k\sin a = \sqrt{3}$ ✓ $\bullet^2$
$\tan a = \sqrt{3}$ so $a = \frac{\pi}{3}$ ✓ $\bullet^4$

Candidate G

$k\cos A\cos B - k\sin A\sin B$ ✗ $\bullet^1$
$k\cos x = 1$ ✗ $\bullet^2$
$k\sin x = \sqrt{3}$
$\tan x = \sqrt{3}$ so $x = \frac{\pi}{3}$ ✓ $\bullet^4$

Candidate H

$k\cos A\cos B - k\sin A\sin B$ ✗ $\bullet^1$
$k\cos B = 1$
$k\sin B = \sqrt{3}$ ✓ $\bullet^2$
$\tan B = \sqrt{3}$ so $B = \frac{\pi}{3}$ ✓ $\bullet^4$

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22 (b) Find the points of intersection of the graph of $y = \cos x - \sqrt{3}\sin x$ with the x and y axes, in the interval $0 \le x \le 2\pi$. **3**

Generic Scheme

22 (b)

- $\bullet^5$ ic interpret y-intercept
- $\bullet^6$ ss strategy for finding roots
- $\bullet^7$ ic state both roots

Illustrative Scheme

- $\bullet^5$ 1
- $\bullet^6$ e.g. $2\cos\left(x+\frac{\pi}{3}\right)=0$ **or** $\sqrt{3}\sin x = \cos x$
- $\bullet^7$ $\frac{\pi}{6}, \frac{7\pi}{6}$

3

Notes

8. Candidates should only be penalised once for leaving their answer in degrees in (a) and (b).
9. If the expression used in (b) is not consistent with (a) then only $\bullet^5$ and $\bullet^7$ are available.
10. Correct roots without working cannot gain $\bullet^6$ but will gain $\bullet^7$.
11. Candidates should only be penalised once for not simplifying $\sqrt{4}$ in (a) and (b).

Regularly occurring responses

Response 4 : Communication for $\bullet^5$

Candidate I

(1, 0) without working. ✗ $\bullet^5$

Candidate J

$\cos 0 - \sqrt{3}\sin 0 = 1$ ✓ $\bullet^5$

so (1, 0).

Response 5 : Follow through from a wrong value of a

Candidate K

From (a) $a = \frac{\pi}{6}$ $\bullet^6$ ✗

then in (b) $x = \frac{\pi}{3}, \frac{4\pi}{3}$ only $\bullet^7$ ✓

Candidate L

From (a) $a = 60^\circ$ ✗ $\bullet^4$ $\bullet^6$ ✗

then in (b) $x = 30^\circ, 210^\circ$ only $\bullet^7$ ✓

Note 10

Response 6 : Root or graphical approach

Candidate M

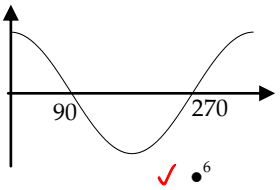
$\frac{\pi}{2} - \frac{\pi}{3}$ and $\frac{3\pi}{2} - \frac{\pi}{3}$ ✓ $\bullet^6$

$= \frac{\pi}{6}$ and $\frac{7\pi}{6}$ ✓ $\bullet^7$

Candidate N

(a) 60° ✗ $\bullet^4$

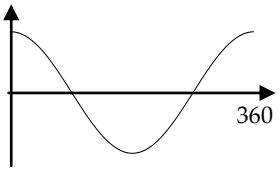
(b)



✓ $\bullet^6$

When $x = 30^\circ, 210^\circ$ ✗ $\bullet^7$

Candidate O



moved 60° to left ✓ $\bullet^6$

cuts x-axis at $\frac{\pi}{6}, \frac{2\pi}{3}$ ✗ $\bullet^7$

Response 7 : Circular argument not leading anywhere

Candidate P

$2\cos x \times \frac{1}{2} - 2\sin x \times \frac{\sqrt{3}}{2} = 0$ $\bullet^6$ ✗

$\cos x - \sqrt{3}\sin x = 0$ $\bullet^7$ ✗

Response 8 : Transcription error in (b)

Candidate Q

(a) correct ✗ $\bullet^6$

(b) $2\cos\left(x-\frac{\pi}{3}\right)=0$ so $x = \frac{5\pi}{6}, \frac{11\pi}{6}$ ✗ $\bullet^7$

$y = 2\cos\left(0-\frac{\pi}{3}\right) = 2\cos\left(-\frac{\pi}{3}\right) = 1$ ✗ $\bullet^5$

$x - \frac{\pi}{3}$ is penalised as $x + \frac{\pi}{3}$ obtained in (a).

However $\bullet^5$ and $\bullet^7$ are still available as follow through. See Note 9.

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23 (a) Find the equation of ℓ_1, the perpendicular bisector of the line joining P(3, –3) to Q(–1, 9). **4**

Generic Scheme

23 (a)

- $\bullet^1$ ss find midpoint of PQ
- $\bullet^2$ ss find gradient of PQ
- $\bullet^3$ ic interpret perpendicular gradient
- $\bullet^4$ ic state equation of perp. bisector

Illustrative Scheme

- $\bullet^1$ (1, 3)
- $\bullet^2$ –3
- $\bullet^3$ $\frac{1}{3}$
- $\bullet^4$ $y-3=\frac{1}{3}(x-1)$

4

Notes

1. $\bullet^4$ is only available if a midpoint **and** a perpendicular gradient are used.
2. Candidates who use $y=mx+c$ must obtain a numerical value for c before $\bullet^4$ is available.

Regularly occurring responses

Response 1 : Candidates who use wrong midpoint or no midpoint

Candidate A

midpoint M(2, –6) ✗ — ✗ $\bullet^1$

$m_{MQ}=-5$ ✓ — ✓ $\bullet^2$

$m_{\perp}=\frac{1}{5}$ ✓ — ✓ $\bullet^3$

$y-(-6)=\frac{1}{5}(x-2)$ ✓ — ✓ $\bullet^4$

Candidate B

$m_{PQ}=-3$ ✓ — ✗ $\bullet^1$

$m_{\perp}=\frac{1}{3}$ ✓ — ✓ $\bullet^2$

using R, $y-(-2)=\frac{1}{3}(x-1)$ ✗ — ✓ $\bullet^3$

✗ $\bullet^4$

23 (b) Find the equation of ℓ_2 which is parallel to PQ and passes through R(1, –2). 2

Generic Scheme

23 (b)

- $\bullet^5$ ic use parallel gradients
- $\bullet^6$ ic state equation of line

Illustrative Scheme

- $\bullet^5$ –3 **stated, or implied by** $\bullet^6$
- $\bullet^6$ $y-(-2)=-3(x-1)$

2

Notes

3. $\bullet^6$ is only available to candidates who use R and their gradient of PQ from (a).

Regularly occurring responses

Response 2 : Not using parallel gradient for equation

Candidate C

$y-(-2)=\frac{1}{3}(x-1)$ ✗

$\bullet^5$ ✗

$\bullet^6$ ✗

Candidate D

Parallel so same gradients

so $m=\frac{1}{3}$ ✗

$y-(-2)=\frac{1}{3}(x-1)$

$\bullet^5$ ✗

$\bullet^6$ ✗

Candidate E

$m=-3$ ✓

$y-(-2)=\frac{1}{3}(x-1)$ ✗

$\bullet^5$ ✓

$\bullet^6$ ✗

If $m_{PQ}=-3$ only do not award $\bullet^5$

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23 (c)	Find the point of intersection of ℓ_1 and ℓ_2.	3

Generic Scheme | **Illustrative Scheme**

23 (c)

$\bullet^7$	ss	use valid approach	$\bullet^7$	e.g. $x-3y=-8$ **and** $9x+3y=3$ **or** $-3x+1=\frac{1}{3}x+\frac{8}{3}$ **or** $3(3y-8)+y=1$
$\bullet^8$	pd	solve for one variable	$\bullet^8$	e.g. $x=-\frac{1}{2}$
$\bullet^9$	pd	solve for other variable	$\bullet^9$	e.g. $y=\frac{5}{2}$

3

Notes

4. **Neither** $x-3y=-8$ and $3x+y=1$ **nor** $y=-3x+1$ and $3y=x+8$ are sufficient to gain $\bullet^7$.
5. $\bullet^7$, $\bullet^8$ and $\bullet^9$ are not available to candidates who:
   - Equate zeros
   - Give answers only, without working
   - Use R for equations in both (a) and (b)
   - Use the same gradient for the lines in (a) and (b).

23 (d)	Hence find the shortest distance between PQ and ℓ_2.	2

Generic Scheme | **Illustrative Scheme**

23 (d)

$\bullet^{10}$	ss	identify appropriate points	$\bullet^{10}$	(1, 3) and $\left(-\frac{1}{2}, \frac{5}{2}\right)$
$\bullet^{11}$	pd	calculate distance	$\bullet^{11}$	$\sqrt{\frac{5}{2}}$ accept $\frac{\sqrt{10}}{2}$ or $\sqrt{2\cdot5}$

2

Notes

6. $\bullet^{10}$ and $\bullet^{11}$ are only available for considering the distance between the midpoint of PQ and the candidate's answer from (c) **or** for considering the perpendicular distance from P or Q to ℓ_2.
7. At least one coordinate at $\bullet^{10}$ stage must be a fraction for $\bullet^{11}$ to be available.
8. There should only be one calculation of a distance to gain $\bullet^{11}$.

Regularly occurring responses

Response 3 : Following through from correct (a), (b) and (c)

Candidate F

(1, 3), (1, −2) X $\bullet^{10}$

d = 5 ✓ $\bullet^{11}$

Response 4 : Following through from correct (a), (b) and (c)

Candidate G

(1, 3), $\left(-\frac{1}{2}, \frac{5}{2}\right)$ ✓ $\bullet^{10}$

PR = $\sqrt{5}$, QR=$\sqrt{125}$, d = $\sqrt{2\cdot5}$

so $\sqrt{2\cdot5}$ is shortest distance. X $\bullet^{11}$

If reference was made to this being the perpendicular distance then $\bullet^{11}$ would be available.

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